浙江大学 2024－2025 学年春夏学期

《离散数学及其应用》课程期末考试试卷

课程号：__211B0010__，开课学院：__计算机学院__

考试试卷：√A 卷、B 卷（请在选定项上打 √）

考试形式：√闭、开卷（请在选定项上打 √），允许带__/__入场

考试日期：__2025__年__6__月__17__日，考试时间：__120__分钟

诚信考试，沉着应考，杜绝违纪。

考生姓名：_____学号：_____任课老师：_____所属院系：_____

题序	一	二	三	四	五	六	七	八	总 分
得分									
评卷人									

1. (18 marks, 2 marks each) Determine whether the following statements are true or false. If it is true write a √ otherwise a × in the blank before the statement.

1) (F) The function $f : Z \times Z \to Z$ where $f(n, m) = 5n + 3m$ is onto and one-to-one.

2) (T) There exist $m \geq 2$ and $n \geq 1$ such that there are more non-isomorphic full m-ary trees with n nodes than n+1 nodes.

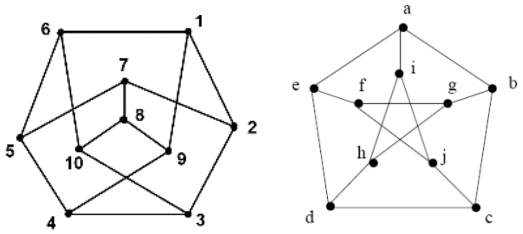
3) (F) The union of two equivalence relations is also an equivalence relation.

4) (F) The worst-case time complexity of Dijkstra's algorithm for computing the shortest path between two points in a weighted graph with n vertices and m edges is $O(n*m)$.

5) (T) If A, B and C are sets, then $(A - B) \cup (A - C) = A \Leftrightarrow A \cap B \cap C = \emptyset$.

6) (T) The quadratic equation $ax^2 + bx + c = 0$ has no rational solutions when a, b, c are odd integers.

7) (T) The given pair of graphs as shown below is isomorphic.



8) (F) $\binom{0.5}{3} \times \binom{0.5}{4} > 0$

9) (T) Suppose that G is a bipartite graph with bipartition (V1, V2), then the sums of the degrees of vertices in the two partitions (V1, V2) are equal.

2. (12 marks, 2 marks each) Multiple-Choice Questions (One or more answers may be correct. Choose all that apply.)

1. (C) For any relation T on a set, let t(T) be its transitive closure and s(T) its symmetric closure. Now, let R and S be relations on a set A. Which of the following relations are true?

(A) $t(R \cup S) = t(R) \cup t(S)$ (B) $t(R \cap S) = t(R) \cap t(S)$

(C) $s(R \cup S) = s(R) \cup s(S)$ (D) $s(R \cap S) = s(R) \cap s(S)$

2. (A, C) Which of the following statements are true? (Assume the same domain is used for all variables.)

(A) $\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x)$

(B) $\exists x(P(x) \wedge Q(x)) \equiv \exists xP(x) \wedge \exists xQ(x)$

(C) $\neg\exists x\,(P(x) \wedge Q(x)) \equiv \forall x\,(\neg P(x) \vee \neg Q(x))$

(D) $\exists x(P(x) \rightarrow Q(x)) \equiv \exists xP(x) \rightarrow \exists xQ(x)$

3. (C) Use Huffman coding to encode these letters with the given frequencies: A:0.10, B:0.22, C:0.05, D:0.15, E:0.26, F:0.07, G:0.06, I:0.09. What is the average number of bits required to encode a symbol?

(A) 2.61 (B) 2.65 (C) 2.79 (D) 2.83

4. (AC) Suppose $f : R \rightarrow R$ is a function. For every $x, y \in R$, $f(x + y) = f(x) + f(y)$.

Which of the following statements are true?

(A) $f(0) = 0$

(B) $f(1) = 0$

(C) for every $n \in N$, $f(n) = n \cdot f(1)$

(D) None of above

5. (A) How many non-isomorphic unrooted spanning trees are there in the wheel W4 ?

(A) 3 (B) 5 (C) 6 (D) 8

6．(B) Let S= {x∈N ∣ 1≤x≤2025}. How many numbers at most can be selected from S to form a subset T ⊆ S, such that for any two different numbers x, y∈T, their sum is not divisible by their absolute difference.

(A) 1012 (B) 675 (C) 506 (D) 405

3. (30 marks, 3 marks each blank) Filling in the blanks.

1) Find the full conjunctive normal form of $(p \oplus q) \rightarrow r$.

$(p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r)$

2) For which m, n such that $K_{m,n}$ is planar? (m<=n) $m \leq 2$

3) How many positive divisors (正约数) does 2025 have 15

4) We can measure the connectivity of a connected graph G = (V, E) in terms of the minimum number of edges that we can remove to disconnect it. If a graph has a cut edge, then we need only remove it to disconnect G. If G does not have a cut edge, we look for the smallest set of edges that can be removed to disconnect it. A set of edges E' is called an **edge cut** of G if the subgraph $G - E'$ is disconnected. The **edge connectivity** of a graph G, denoted by λ **(G)**, is the minimum number of edges in an edge cut of G.

Find λ ($K_{m,n}$), where m and n are positive integers: $min(m,n)$

5) Find the coefficient of x^5 in the expansion of $\left(x - \frac{2}{x}\right)^9$ 144

6) There are three consecutive positive integers that are divisible by 7, 5, and 11 respectively (三个连续的正整数，依次能被7, 5, 11整除). Find the smallest positive integer solution. 119,120,121

7) A labeled tree is a tree where each vertex is assigned a label. Two labeled trees are considered isomorphic when there is an isomorphism between them that preserves the labels of vertices. For example, there are 3 non-isomorphic trees with 3 vertices labeled with different integers from the set {1, 2, 3}. How many non-isomorphic trees are there with four vertices labeled with different integers from the set {1, 2, 3, 4}? 16

8) Write the first 6 terms of the sequence determined by the generating function: (1-2x)/(1+2x).

1,-4,8,-16,32,-64

9） How many ways are there to distribute 6 distinct apples to John, Alice, Tom and Sunny, such that exactly one person receives no apples? 2160

10) If A, B and C are sets, A ∪ B ∪ C={1,2,3,4,5,6}, A ∩ B ={5}, and B ∩ C={2}. Find the number of all the possible cases of the sets A, B and C. 256

4. (8 marks) For the poset ({2, 3, 5, 6, 12, 20, 27, 60, 108}, |),

1） Draw the Hasse diagram. (2 marks)

2） Find the maximal elements. (1 mark)

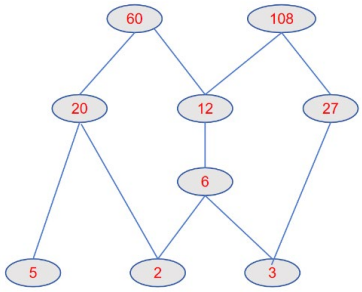
3） Find the least element. (1 mark)

4） Find all upper bounds of {3, 5}(1 mark)

5） Determine whether the underlying undirected graph of the Hasse diagram has a Hamilton path. (1 mark)

6） Use the topological sorting to order the elements of the poset. If there are multiple solutions, choose the lexicographically smallest sequence. (2 marks)

(1)



（2）60，108；

（3）Does not exist；

（4）60；

（5）This graph has a Hamilton path: 5，20，2，6，3，27，108，12，60；

（6）2，3，5，6，12，20，27，60，108。

5. (5 marks) Given directed graph *G* in Fig.1.

(1) Find the strongly connected components of *G*. (2 marks)

(2) How many paths of length 4 are there in *G*? How many of them are cycles? (3 marks)

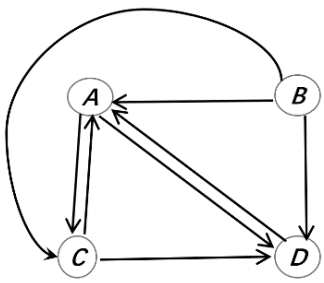


Fig. 1

(1) {A，C，D}、{B}；

(2) The adjacency matrix of G is

$$M = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, $M^4 = \begin{bmatrix} 4 & 0 & 1 & 3 \\ 5 & 0 & 3 & 5 \\ 3 & 0 & 2 & 3 \\ 1 & 0 & 2 & 2 \end{bmatrix}$

The number of paths of length 4 is 34, and the number of cycles among them is 8.

6. (7 marks) For a connected simple graph, use these standard definitions:

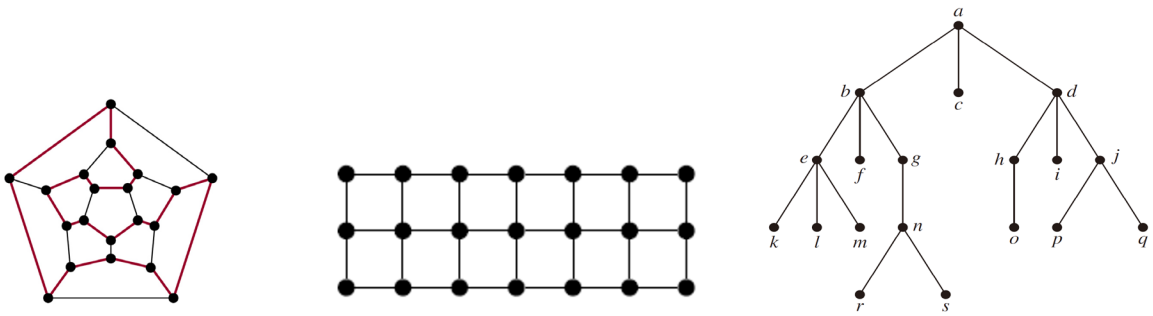
- **Distance** between two distinct vertices v_1 and v_2: Length (number of edges) of the shortest path between them.
- **Eccentricity** of a vertex v: Maximum distance from v to any other vertex.
- **Radius, $r(G)$**: Minimum eccentricity over all vertices.
- **Diameter, $d(G)$**: Maximum eccentricity over all vertices
- **Center**: The set of vertices whose eccentricity equals the $r(G)$. These vertices are called **central vertices**.

(1) Find $r(G)$ in Fig.2(a). (1 mark)

(2) Find $d(G)$ in Fig.2(b). (1 mark)

(3) List all vertices in the center in Fig.2(c). (2 marks)

(4) For any connected simple graph G, determine whether $r(G) \leq d(G) \leq 2r(G)$ is always true, and **explain why**. (3 marks)



(a) (b) (c)

Fig. 2

（1）5； （2）8;（3）a,b;

（4）The statement is true.

By definition of eccentricity，$r(G) \leq d(G)$.

let (x, y) denote the distance from x to y.

Since the graph is simple and connected, for any two vertices x, y, we have

$$(x, y) \leq (x, u) + (u, y) \leq r(G) + r(G) = 2r(G)$$

Therefore, $d(G) \leq 2r(G)$, completing the proof.

7. (10 marks) An undirected simple graph is **regular** if every vertex has the same degree. It is *k*-**regular** if every vertex has degree *k*.

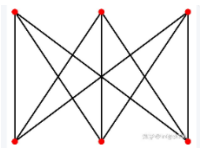
1） Prove that every 2-regular graph is a disjoint union of cycles (不相交的环的并). (3 marks)

2） Draw **two** non-isomorphic 3-regular graphs, each with 6 vertices. (3 marks)

3） For each graph from part (2), determine whether it is planar. (2 marks)

4） For each graph from part (2), find its chromatic number (着色数). (2 marks)

（1）Starting from any vertex and walking along both incident edges, one eventually returns to the starting vertex, since each vertex has degree 2. Therefore, a connected 2-regular graph is a cycle. If the graph is disconnected, then each 2-regular component is an independent cycle.

（2）  

（3）$K_{3,3}$ is nonplanar，the triangular prism is planar；

（4）$\chi(K_{3,3})$=2，$\chi(triangular\ prism)$=3.

8. (10 marks) Let G_n be the graph consisting of a path with n vertices together with a new vertex adjacent to each of these n vertices, as shown in Fig.3. So G_n has exactly **n+1** vertices.

Let a_n denote the number of spanning trees in G_n with a_1 =1.

1）Calculate a_2 (2 mark)

2）Solve the recurrence formula for a_n :

$a_n = 3a_{n-1} - a_{n-2}$ $(n \geq 3)$ (4 marks)

3）Given Fibonacci numbers f_n with f_1=1and f_2=1, use induction to show that $a_n = f_{2n}$. (4 marks)

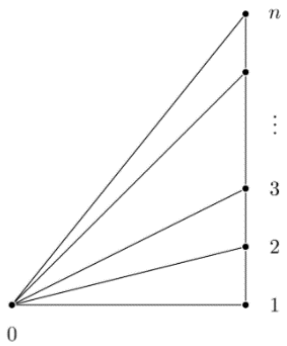


Fig. 3

（1）$a_2 = 3$;

（2）$a_n = \frac{1}{\sqrt{5}}\left[\left(\frac{3+\sqrt{5}}{2}\right)^n - \left(\frac{3-\sqrt{5}}{2}\right)^n\right]$

（3）Basis step: $f_2 = a_1, f_4 = a_2$

Inductive step: Assume $a_k = f_{2k}$, for $k \leq n$

$k = n + 1$: $f_{2n+2} = f_{2n+1} + f_{2n} = 2f_{2n} + f_{2n-1} = 3f_{2n} - f_{2n-2} = a_{n+1}$

By strong induction, $a_n = f_{2n}$.